input: $n, d \in \mathbb{N}$, $\mathsf{LDC} : \{0,1\}^r \to \{0,1\}^R$, and stream $z \in \{0,1\}^{R^d}$.

function recurse_linear(integer $a \le d$, stream $z \in \{0,1\}^{N_{inner} R^a}$, tuple $T \in [r]^{d-a}$):

- **if** $a = 0$ **then**
  - Set σ to the decoding of z via C_{inner} with probability $1 - 2\delta(z, \mathsf{C}_{inner}(\sigma))$ and otherwise to $\perp$. This is the only step where the stream is read, and it happens in unison for all instances reading this stream; the randomness in deciding σ is the same for all of them.
  - **return** $\ell_T \cdot \sigma$ ($\perp$ if $\sigma = \perp$).
- Pick lists $\mathsf{qlist}_1, \ldots, \mathsf{qlist}_r$ as in Lemma 16.
- Define maps $\mathcal{M}_i : \mathsf{qlist}_i \to \mathbb{K}$ for $i \in [r]$ as follows:
- **for** $I \in [R]$ **do**
  - Let z_I denote the next R^{a-1} elements of the stream z.
  - **for** i such that $I \in \mathsf{qlist}_i$ **do**
    - Set $\mathcal{M}_i[I] \leftarrow$ recurse_linear$(a-1, z_I, T|i)$.
- **for** $i \in [r]$ **do**
  - Compute $p_i(\sigma)$ for all $\sigma \in \mathbb{K}$ using local decoding on index i with queries $\mathcal{M}_i$.
  - Set the guess $\sigma_i = \arg\max_\sigma p_i(\sigma) \in \mathbb{K}$ and likelihood $p_i = \max_\sigma p_i(\sigma)$.
- Let $p = \min_i p_i$.
- **return** $\sigma_1 + \ldots + \sigma_r$ with probability p and otherwise $\perp$.

Compute recurse_linear$(d, z, ())$ in parallel $(\log n)^2$ times. When the output is $\perp$, assign it 0 or 1 randomly and otherwise project to $\mathbb{F}_2$. Let majority output be σ.

output: σ.